

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022**(CBCSS-UG)****Mathematics****MTS 6B 12--CALCULUS OF MULTIVARIABLE****(2019 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 6B 12—CALCULUS OF MULTIVARIABLE

(Multiple Choice Questions for SDE Candidates)

- If $f(x, y) \geq g(x, y)$ on D , then $\iint_D f(x, y) dA \dots$
 - $\leq \iint_D g(x, y) dA.$
 - $\geq \iint_D g(x, y) dA.$
 - 0.
 - None of these.
- Evaluate $\iint_R (1 - 2xy^2) dA$, where $R = \{(x, y) \mid 0 \leq x \leq 2, -1 \leq y \leq 1\}$:
 - $\frac{4}{3}.$
 - $\frac{1}{3}$
 - 1.
 - 0.
- Evaluate $\iiint_T x dV$, where T is the part of the region in the first octant lying inside the sphere $x^2 + y^2 + z^2 = 1$.
 - $\frac{\pi}{6}.$
 - $\frac{\pi}{16}.$
 - $\frac{\pi}{4}.$
 - $\pi.$
- Evaluate $\lim_{(x,y) \rightarrow (1,2)} (x^3 y^2 - x^2 y + x^2 - 2x + 3y)$:
 - 0.
 - 1.
 - 7.
 - 7.
- Let f be a function that is defined for all points (x, y) close to the point (a, b) . Then f is continuous at the point (a, b) if :
 - $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$
 - $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists.
 - $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ does not exist.
 - $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(0, 0).$
- A region R is said to be an open region if every point of R is a/an _____.
 - Cluster point.
 - Interior point of R .
 - Closure of R .
 - None of these.
- Determine where the function is continuous : $f(x, y) = \frac{xy(x^2 - y^2)}{x^2 + y^2}$:
 - Everywhere except at $(0, 0)$.
 - Everywhere.
 - Only at the center.
 - At $(1, 1)$.

8. Evaluate $\lim_{(x,y,z) \rightarrow \left(\frac{\pi}{2}, 0, 1\right)} \frac{e^{2y}(\sin x + \cos y)}{1 + y^2 + z^2}$

- (A) 0. (B) 1.
(C) 2. (D) -2.

9. Find $\frac{\partial f}{\partial x}$ if $f(x, y) = 2x^2y^3 - 3xy^2 + 2x^2 + 3y^2 + 1$:

- (A) $6x^2y^3 - 3xy^2 + 4x^2 + 3y^2$. (B) $4xy^3 - 3y^2 + 4x + 3y^2$.
(C) $6x^2y^2 - 6xy + 6y$. (D) 0.

10. Suppose z is a differentiable function of x and y that is defined implicitly by $x^2 + y^3 - z + 2yz^2 = 5$. Find $\partial z / \partial x$.

- (A) $\frac{2}{1-4yz}$. (B) $\frac{2x}{1-4yz}$.
(C) $\frac{2x}{4yz}$. (D) None of these.

11. The function $(x, y) = e^x \cos y$ is _____ in the xy -plane.

- (A) Real. (B) Harmonic.
(C) Conjugate. (D) Complex.

12. Let $w = 2x^2y$, where $x = u^2 + v^2$ and $y = u^2 - v^2$. Find $\partial w / \partial v$.

- (A) $4xu(2y + x)$. (B) $2xu(2y + x)$.
(C) $4xu(2y - x)$. (D) $2xv(2y + x)$.

13. Find $\frac{\partial z}{\partial y}$ if $2x^2z - 3xy^2 + yz - 8 = 0$.

- (A) $\frac{6xy - z}{2x^2 + y}$. (B) $\frac{3y^2 - 4xz}{2x^2 + y}$.
(C) $-\frac{3x^2 + y}{x + 2y}$. (D) $-\frac{3x^2 + z}{x + 2y}$.

14. Compute $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ given that $\mathbf{F}(x, y, z) = (x + \sin z)\mathbf{i} + (2y + \cos x)\mathbf{j} + (3z + \tan y)\mathbf{k}$ and S is the unit sphere $x^2 + y^2 + z^2 = 1$.
- (A) 8π . (B) 4π .
(C) 2π . (D) π .
15. If $\nabla f(x, y) = 0$, then $D_u f(x, y)$ ————— for every u .
- (A) $= 0$. (B) > 0 .
(C) < 0 . (D) $= 1$.
16. Let f be continuous on a polar rectangle $R = \{(r, \theta) \mid 0 \leq a \leq r, \alpha \leq \theta \leq \beta\}$, where $0 \leq \beta - \alpha \leq 2\pi$. Then $\iint_R f(x, y) dA =$ —————.
- (A) $\int_a^\beta \int_a^b f(\cos \theta, \sin \theta) r dr d\theta$. (B) $\int_a^\beta \int_a^b f(r \cos \theta, r \sin \theta) dr d\theta$.
(C) $\int_a^\beta \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$. (D) $-\int_a^\beta \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$.
17. The Jacobian of the transformation T defined by $x = g(u, v)$ and $y = h(u, v)$ is :
- (A) $\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$. (B) $\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$.
(C) $-\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$. (D) $-\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} + \frac{\partial y}{\partial u} \frac{\partial x}{\partial v}$.
18. If $w = f(x^2 - y^2, y^2 - x^2)$ and f is differentiable, then $y \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial y} =$ —————.
- (A) 2. (B) 0.
(C) -1. (D) 1.
19. Let f be a function of two variables x and y . The gradient of f is the vector function $\nabla f(x, y) =$ —————.
- (A) $f_y(x, y)\mathbf{i} + f_x(x, y)\mathbf{j}$. (B) $f_x(x, y)\mathbf{j} + f_y(x, y)\mathbf{i}$.
(C) $f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$. (D) $f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{k}$.
20. Suppose that f has continuous second-order partial derivatives on an open region containing a critical point (a, b) of f . Let $D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - f_{xy}^2(x, y)$. If $D(a, b) > 0$ and $f_{ww}(a, b) < 0$, then $f(a, b)$ is a —————.
- (A) Relative minimum value. (B) Saddle point.
(C) Relative maximum value. (D) None of these.

SIXTH SEMESTER U.G. DEGREE EXAMINATION, MARCH 2022

(CBCSS-UG)

Mathematics

MTS 6B 12—CALCULUS OF MULTIVARIABLE

(2019 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A (Short Answer Questions)

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find the domain and range of the function $f(x, y) = x + 3y - 1$.
2. Evaluate $\lim_{(x, y, z) \rightarrow \left(\frac{\pi}{2}, 0, 1\right)} \frac{e^{2y} (\sin x + \cos y)}{1 + y^2 + z^2}$.
3. Find f_x and f_y if $f(x, y) = x \cos xy^2$.
4. Find the directional derivative of $f(x, y) = x^2 \sin 2y$ at $\left(1, \frac{\pi}{2}\right)$ in the direction of $\vec{v} = 3\hat{i} - 4\hat{j}$.
5. Find $\nabla f(x, y, z)$ if $f(x, y, z) = x^2 + y^2 - 4z$ and find the direction of maximum increase of f at the point $(2, -1, 1)$.
6. Find the relative extrema of the function $f(x, y) = x^2 + y^2 - 2x + 4y$.
7. Find the volume of the solid lying under the elliptic paraboloid $z = 8 - 2x^2 - y^2$ above the rectangular region given by $0 \leq x \leq 1, 0 \leq y \leq 2$.
8. Find the mass of the triangular lamina with vertices $(0, 0), (0, 3)$ and $(2, 3)$ given that the density at (x, y) is $\rho(x, y) = 2x + y$.

Turn over

9. Find the Jacobian for the change of variables defined by $x = r \cos \theta, y = r \sin \theta$.
10. Evaluate $\iint_R 2x - y \, dA$ where R is the region bounded by the parabola $x = y^2$ and the line $x - y = 2$.
11. Find whether the vector field $\vec{F} = x^2y \hat{i} + xy \hat{j}$ is conservative.
12. State Green's theorem.
13. Find a parametric representation for the cone $x = \sqrt{y^2 + z^2}$.
14. Find the surface area of the torus given by $\vec{r}(u, v) = (2 + \cos u) \cos v \hat{i} + (2 + \cos u) \sin v \hat{j} + \sin u \hat{k}$ where the Domain D is given by $0 \leq u \leq 2\pi$ and $0 \leq v \leq 2\pi$.
15. Compute $\iint_S \vec{F} \cdot \hat{n} \, dS$ given $\vec{F}(x, y, z) = (x + \sin z) \hat{i} + (2y + \cos x) \hat{j} + (3z + \tan y) \hat{k}$ and S is the unit sphere $x^2 + y^2 + z^2 = 1$.

(10 × 3 = 30 marks)

Section B (Paragraph Questions)

*Answer at least five questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 30.*

16. Find f_{xyx} and f_{yx} if $f(x, y) = x \cos y + y \sin x$.
17. Find the differential of $w = x^2 + xy + z^2$. Compute the value of dw if (x, y, z) changes from $(1, 2, 1)$ to $(0.98, 2.03, 1.01)$ and compare the value with that of Δw .
18. Find the equation of the tangent plane and normal line to the surface $x^2 - 2y^2 - 4z^2 = 4$ at $(4, -2, -1)$.
19. Find the relative extrema of $f(x, y) = -x^3 + 4xy - 2y^2 + 1$.
20. Find the volume of the solid region bounded by the paraboloid $z = 4 - x^2 - 2y^2$ and the xy -plane.

21. Evaluate $\int_0^2 \int_0^x \int_0^{x+y} e^x (y + 2z) dz dy dx$.

22. Determine whether the vector field $\vec{F} = e^x (\cos y \hat{i} - \sin y \hat{j})$ is conservative. If so, find a potential function for the vector field.

23. Evaluate $\oint_C (e^x + y^2) dx + (x^2 + 3xy) dy$, where C is the positively oriented closed curve lying on the boundary of the semi annular region R bounded by the upper semicircles $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$ and the x-axis.

(5 × 6 = 30 marks)

Section C (Essay Questions)

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Sketch the graph of $f(x, y) = 9 - x^2 - y^2$.

(b) Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$ does not exist.

25. (a) Find the relative extrema of $f(x, y) = x^3 + y^2 - 2xy + 7x - 8y + 2$.

(b) Find the minimum value of $f(x, y, z) = 2x^2 + y^2 + 3z^2$ subject to the constraint $2x - 3y - 4z = 49$.

26. Let R be the region bounded by the square with vertices (0,1), (1,2), (2,1) and (1,0). Evaluate

$$\iint_R (x+y)^2 \sin^2(x-y) dA.$$

27. Let $\vec{F} = (x, y, z) = 2xyz^2 \hat{i} + x^2z^2 \hat{j} + 2x^2yz \hat{k}$.

(a) Show that $\vec{F}$ is conservative and find a scalar function f such that $\vec{F} = \nabla f$.

(b) If $\vec{F}$ is a force field, find the work done by $\vec{F}$ in moving a particle along any path from (0, 1, 0) to (1, 2, -1).

(2 × 10 = 20 marks)